# The Coffee Law: Information, Reasoning and Communication Ceilings for Tool-Using Agent Collectives

Manish Bhatt
*Affiliation to be completed*
*email to be completed*

***Abstract*—Why do some multi-agent systems improve with more agents, more debate or a stronger model, while others do not? We argue that the productive question is not how to make a collective more intelligent, but *which resource limits a particular collective, and which intervention removes that limit*. We decompose collective intelligence into three ceilings, each bounded by a different theory. *Information* (information theory): a team cannot know more about the world than its starting data plus what its tools can carry, $\min\{H(\Theta),\, I(\Theta; D_0) + \sum_t C_t\}$ bits; talking and voting do not raise this budget, and adding agents raises it only through more tool access. *Reasoning* (computational complexity): what bounded agents can *use* is at most what they *hold*; debate and compute close the gap but never pass it, at a cost ranging from linear to, under standard cryptographic assumptions, infeasible. *Communication* (communication complexity): evidence split across agents is unusable to any one of them until enough is exchanged, about $n/2$ bits for some $n$-bit tasks even with unlimited computation. Adding agents, adding communication, adding computation and adding access to the world are therefore different interventions acting on different ceilings. The decomposition explains published findings that otherwise look contradictory, yields a diagnostic for which ceiling binds, and makes predictions that a single notion of capability does not, such as that model diversity and prompt diversity help on different tasks. Proofs are short; every finite example is verified exactly in public code.**

***Index Terms*—multi-agent systems, LLM agents, tool use, information theory, channel capacity, usable information, communication complexity**

## I. Introduction

A team of LLM agents is paged about a production incident. Two questions arrive. *Which server is failing?* The answer is in live logs the team has not yet read. *Is this patch to the locking code correct?* The answer is fully determined by code the team already has. A common belief is that adding agents, rounds of debate or thinking time helps with both. It helps with only one of them, and the reason is precise: the two questions hit different ceilings.

The first is limited by **information**. No amount of discussion reveals log lines nobody has fetched; only tool calls (reading logs, running queries) do, and each call can carry only so much. The second is limited by **reasoning**: the answer is already in hand, and the work is computing it. A third limit appears when evidence is split across agents, since what one agent holds is useless to another until it is communicated: **communication**.

Our thesis is that *different mechanisms of collective intelligence have different kinds of bounds*. Information theory constrains what can be learned from observations; computational complexity constrains what can feasibly be derived; communication complexity constrains what restricted exchange can accomplish. Treating these as one question of "model capability" or "agent count" hides which one is binding. Scaling a collective is therefore not a single operation: adding agents, adding communication, adding computation and expanding access to the world act on different ceilings (Fig. 1), and the useful question for a struggling collective is *which resource limits it, and what intervention can remove that limit*.

We distinguish three levels of contribution, and claim them in that order of strength.

- **Conceptual.** The three-ceiling decomposition, with every operation of a collective classified as either computation on what it holds or a channel to the world, and the chain usable $\leq$ available $\leq$ budget.
- **Scientific utility.** The decomposition explains a body of published findings in one picture (Table II), gives a diagnostic for which ceiling binds, and makes predictions that a single capability axis does not (Section IX).
- **Mathematical.** The ingredients are classical: the chain rule, the data-processing inequality (DPI), channel capacity, Fano's inequality [1], usable information [2] and communication complexity [3]. The statements are, to our knowledge, new in this setting: a capacity budget for adaptive multi-agent tool use in which the agents and protocol drop out (Theorem 1); the reasoning gap as an expected KL divergence with compute bounds on closing it (Theorems 2–4); a communication ceiling for collectives (Theorem 5); and a channel account of sandboxing (Theorem 6).

Full proofs and exact numerical checks of every finite example are public.[1]

The theory is agent-agnostic: it applies equally to people, programs and mixed human–AI teams. It bounds learning about the world and using what is learned; it does not cap intelligence, and it says nothing about outcomes the agents themselves bring about.

[1] https://github.com/mbhatt1/coffeelaw

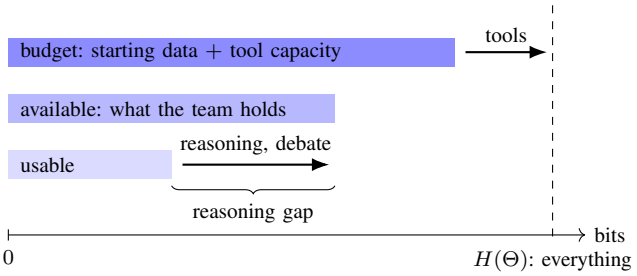



Fig. 1. The ceilings. What a team can use is at most what it holds, which is at most its budget. Tools (and better starting data) raise the budget; reasoning (more compute, debate, more agents) raises usable information toward what is held, never past it. When data is split across agents, a third ceiling, communication, limits what each agent can use.

## II. Model

*Intuition.* We track one object: the *record*, everything the team has ever held. Each round, agents may call tools; tool answers depend on the world; everything else the agents do (posting, debating, voting, thinking) is a function of the record plus coin flips.

All variables are discrete with finite support (Section VII treats continuous ones); logarithms are base 2. $\Theta$ is the truth the team wants. $D_t$ is the record at the start of round $t$: the initial data $D_0$ (every agent's context and model) plus every post, private state, query and tool result since. In round $t$, internal randomness $\xi_t$ is drawn; tool calls $Q_t = g_t(D_t, \xi_t)$ are issued under any policy; results $O_{t+1} \sim K_t(\cdot \mid \Theta, Q_t, D_t)$ return; communication and computation use fresh randomness $\zeta_t$; and $D_{t+1} = (D_t, \xi_t, Q_t, O_{t+1}, \zeta_t)$.

**Assumptions.** (A1) *No side channels*: $\Theta$ reaches the team only through $D_0$ and tool results. (A2) *Coin flips are honest*: $\xi_t, \zeta_t$ are independent of $\Theta$ and of everything before them. (A3) *Tools are channels from the world*: given $(\Theta, Q_t, D_t)$, $O_{t+1}$ has law $K_t$ (tools may be adaptive or stateful). (A4) *The truth is fixed*: the team's actions do not change $\Theta$. (A1) is not an assumption about the agents but the definition of which channels exist; Theorem 6 treats agents that can open new ones, and Section VII relaxes the others.

**Capacity.** For each query $q$ and history $d$, the tool is a channel $\theta \mapsto K_t(\cdot \mid \theta, q, d)$ with capacity $\mathrm{Cap}(K_{t,q,d}) = \max_P I(\Theta; O)$, the most it can ever reveal per call. Let $C_t = \sup_{q,d} \mathrm{Cap}(K_{t,q,d})$. Always $C_t \le \min\{\log|\mathcal{O}_t|, \log|\mathrm{supp}\,\Theta|\}$; a pass/fail test has $C_t \le 1$ bit.

**What is random.** For facts about the world, the randomness is over *possible worlds* $W$: a question $X \perp W$ has answer $\Theta = \theta(W, X)$, and training is a channel $W \to$ corpus $\to M$ to the weights $M$. Then $I(\Theta; X, M) \le I(\Theta; X) + I(W; M)$ and $I(W; M) \le I(W; \text{corpus})$: a model knows at most what its training data carried. In one fixed world, $\Theta$ is a function of $X$ and $M$ carries no further information about it, so a model's factual knowledge appears as *usable* information (Section IV): something it computes from the question.

## III. Ceiling 1: Information (the Coffee Law)

*Intuition.* Picture each tool call as a narrow pipe from the world into the team's shared bucket. Talking, voting and thinking move water between cups; they never add water. Only pipes do, and each pipe has a width.

**Theorem 1** (Coffee Law)**.** *Under (A1)–(A4), with* $B_T = I(\Theta; D_0) + \sum_{t<T} C_t$,

$$I(\Theta; D_T) \le \min\{H(\Theta), B_T\},$$

*and the same holds for every output of the team (answer, vote, board, any agent's state). For any estimator* $\hat{\Theta}$ *and* $|\Theta| \ge 3$, $\Pr(\hat{\Theta} \ne \Theta) \ge (H(\Theta) - \min\{H(\Theta), B_T\} - 1)/\log(|\Theta| - 1)$. *In particular, with no tools,* $I(\Theta; D_T) \le I(\Theta; D_0)$*: a closed team cannot learn anything by talking.*

*Proof.* By the chain rule, a round adds $I(\Theta; \xi_t | D_t) + I(\Theta; Q_t | D_t, \xi_t) + I(\Theta; O_{t+1} | S) + I(\Theta; \zeta_t | S, O_{t+1})$ with $S = (D_t, \xi_t, Q_t)$. The first and last terms are 0 by (A2); the second is 0 because $Q_t$ is a function of $(D_t, \xi_t)$. Given $S = s = (d, x, q)$, (A3) makes $(\Theta, O_{t+1})$ an input–output pair of channel $K_{t,q,d}$ with input law $p(\cdot|s)$, so the third term is at most $\mathrm{Cap}(K_{t,q,d}) \le C_t$. Sum over rounds; also $I \le H$. Outputs are functions of $D_T$ and independent noise (DPI). Fano's inequality gives the error floor. □

Choosing queries cleverly is already accounted for: a policy only selects which channel to use and under which posterior, and capacity is the maximum over both, as feedback cannot raise the capacity of a memoryless channel [4], [5].

**Corollary 1** (What does not help)**.** *Debate, voting, critique and the number of agents do not appear in* $B_T$*. More agents raise the budget only by making more, or more informative, tool calls.*

**Corollary 2** (Calls needed)**.** *If* $\Theta$ *is uniform on* $2^k$ *values* ($k \ge 2$)*,* $I(\Theta; D_0) = 0$ *and* $C_t \le C$ *with* $C > 0$*, then error* $\le \varepsilon$ *requires at least* $((1-\varepsilon)k - 1)/C$ *tool rounds, however many agents run and however they talk.*

**Example: debugging.** A test run returns pass or fail, so it carries at most one bit. If the bad commit is equally likely to be any of $n$, identifying it with error at most $\varepsilon$ needs at least $(1-\varepsilon)\log n - 1$ test runs whatever the team does (Corollary 2); binary search (`git bisect`) uses $\lceil \log n \rceil$. A flaky test that misreports with probability $p$ carries at most $1 - h(p)$ bits, raising the minimum by the factor $1/(1 - h(p))$.

**Exact accounting and tightness.** The proof gives the identity $I(\Theta; D_T) = I(\Theta; D_0) + \sum_t \mathbb{E}_s[I(\Theta; O_{t+1} | S_t = s)]$: total information is the sum of what each call actually revealed. The bound is tight exactly when every call is used at full capacity: almost every posterior is a capacity-achieving input for the chosen tool, and each chosen tool attains $C_t$. For rich tools (a web page), the realized per-call gain is the binding quantity; the capacity form bites for low-capacity tools such as tests, checks and short API answers. Table I shows a case where a good policy meets the bound and a poor one wastes it.

TABLE I
EXACT INFORMATION AFTER TWO TOOL ROUNDS. $\Theta$ IS UNIFORM ON 4 VALUES; EACH CALL RETURNS ONE CHOSEN BIT OF $\Theta$, FLIPPED WITH PROBABILITY 0.1 ($C \approx 0.531$ BITS). BUDGET: $\min\{H(\Theta), 2C\}$.

| Policy | $I(\Theta; D_2)$ | Budget |
|---|---|---|
| Ask bit 0, then bit 1 | 1.0620 | 1.0620 |
| Adaptive, may repeat a bit | 0.9820 | 1.0620 |
| Either, plus a debate step | unchanged | — |
| Ask both bits, noise-free tool | 2.0000 | 2.0000 |

## IV. CEILING 2: REASONING

*Intuition.* Holding information is not the same as using it. The prime factors of a number $N$ are completely determined by $N$, so a team holding $N$ holds all the information about them; finding them may still take enormous computation. Reasoning turns held information into usable information.

**Definition 1** ( [2])**.** A *predictive family* $\mathcal{V}$ is a set of predictors $f$ mapping a record $d$ to a distribution $f[d]$ on $\Theta$, including the prior predictor; it models what the agents can compute. Let $H_{\mathcal{V}}(\Theta|D) = \inf_{f\in\mathcal{V}} \mathbb{E}[-\log f[D](\Theta)]$ and define the *usable information* $I_{\mathcal{V}}(D \to \Theta) = H(\Theta) - H_{\mathcal{V}}(\Theta|D)$.

**Theorem 2** (Reasoning ceiling)**.** *(a)* $0 \le I_{\mathcal{V}}(D \to \Theta) \le I(\Theta; D)$*, with* reasoning gap $I(\Theta; D) - I_{\mathcal{V}}(D \to \Theta) = \inf_{f\in\mathcal{V}} \mathbb{E}_D\, D_{\mathrm{KL}}(p(\cdot|D) \,\|\, f[D])$*. (b) Hence* $I_{\mathcal{V}}(D_T \to \Theta) \le I(\Theta; D_T) \le \min\{H(\Theta), B_T\}$*. (c) Computation can raise usable information but never available information: for some* $\mathcal{V}$ *and deterministic* $g$*,* $I(\Theta; g(D)) = I(\Theta; D)$ *while* $I_{\mathcal{V}}(g(D) \to \Theta) > I_{\mathcal{V}}(D \to \Theta)$*; for every* $g$*,* $I_{\mathcal{V}}(g(D) \to \Theta) \le I(\Theta; D)$*.*

*Proof.* (a) The prior predictor gives $H_{\mathcal{V}} \le H(\Theta)$. For any $f$, $\mathbb{E}[-\log f[D](\Theta)] = H(\Theta|D) + \mathbb{E}_D D_{\mathrm{KL}}(p(\cdot|D)\|f[D])$; take the infimum. (b) Part (a) and Theorem 1. (c) Let two agents hold uniform bits $B_1, B_2$ and $\Theta = B_1 \oplus B_2$, and let $\mathcal{V}$ contain the predictors that read at most one coordinate of the record. Each bit alone is independent of $\Theta$, so usable information is 0, although the pair determines $\Theta$. After one reasoning step $g(D) = (B_1 \oplus B_2, B_2)$, one coordinate equals $\Theta$: usable information is 1 bit, available information unchanged. The last claim is (a) and the DPI. □

**Corollary 3** (Debate adds reasoning, not facts)**.** *In a closed team every post is a function of* $D_0$ *and noise, so it adds no available information; it can add usable information, as when an agent posts an intermediate result (here* $B_1 \oplus B_2$*) that others could not compute. The gain is capped by what the team holds.*

The incident example is now exact. "Which server is failing?" is *information-bound*: the right inequality of the chain binds, and only tools help. "Is this patch correct?" is *compute-bound*: the left gap is large, and reasoning helps. A code interpreter run on data already held is reasoning (it enlarges $\mathcal{V}$); a tool that observes the world is information (it raises $C_t$). In Table I's noisy example, the record holds 1.062 bits but a one-look agent can use only 0.266; the rest is a reasoning gap that more observations cannot close.

**Imperfect agents.** Real agents fall further below the chain (proofs in the accompanying notes, footnote 1). Acting on a belief $\hat{b}$ instead of the correct posterior $b^*$ costs at most

$$2U\,\mathrm{TV}(b^*, \hat{b}) + \delta \;\le\; 2U\sqrt{(\ln 2/2)\, D_{\mathrm{KL}}(b^* \| \hat{b})} + \delta$$

for utility range $U$ and own suboptimality $\delta$. A summary $M$ of data $D$ loses exactly $\mathbb{E}\, D_{\mathrm{KL}}(p(\cdot|D)\|p(\cdot|M)) = I(\Theta; D) - I(\Theta; M)$. And a confident message carrying zero information can flip a correct agent: with belief $(0.6, 0.4)$ for the truth, giving weight $\alpha > 1/6$ to a constant false claim flips the answer.

## V. PAYING FOR REASONING

*Intuition.* Closing the reasoning gap costs computation. A team of agents is just a computation spread over several machines, so what counts is total work; and some tasks need a lot of it.

Let $\mathcal{V}_S$ be the predictors computable and samplable in at most $S$ steps of a fixed machine model (reading one input symbol per step). A team's *total compute* is the sum of all agents' steps over all rounds, counting board reads and writes.

**Theorem 3** (Only total compute counts)**.** *A team with total compute* $S$ *outputs a predictor in* $\mathcal{V}_{\kappa S}$ *for a constant* $\kappa$ *of the machine model, so its usable information is at most* $I_{\mathcal{V}_{\kappa S}}(D \to \Theta) \le I(\Theta; D)$*.*

*Proof.* One machine can run the agents step by step, keeping their memories and the board in separate regions, at $O(1)$ cost per simulated step. □

The comparison holds in any fixed cost model (RAM steps, FLOPs, tokens processed), up to the overhead of interleaving.

**Lemma 1.** *Sampling* $\hat{\Theta} \sim f[D]$ *is correct with probability at least* $2^{-\mathbb{E}[-\log f[D](\Theta)]}$ *(Jensen). If* $H(\Theta) \le 1$ *and* $f$ *attains usable information* $\delta$ *(cross-entropy* $H(\Theta) - \delta$*), the success probability is at least* $\frac{1}{2} + \frac{\ln 2}{2}\delta$*.*

**Theorem 4** (Some gaps are expensive)**.** *(a) For the parity* $\Theta$ *of* $n$ *uniform bits, any predictor or team that reads at most* $n-1$ *bits, even adaptively, has zero usable information; reading all* $n$ *gives one bit. (b) If one-way permutations exist, then for the Goldreich–Levin bit* $\Theta = \langle x, r\rangle \bmod 2$ *given* $D = (f(x), r)$*,* $I(\Theta; D) = H(\Theta)$*, yet every polynomial-time team has usable information at most* $\log(1 + 2\varepsilon(n))$ *with* $\varepsilon$ *negligible.*

*Proof.* (a) Given any transcript of fewer than $n$ reads, the unread bits are uniform, so $\Theta$ is independent of the transcript; posts are functions of their authors' reads. (b) Every polynomial-time algorithm guesses $\Theta$ with probability at most $\frac{1}{2} + \varepsilon$ [6]; apply Lemma 1; by Theorem 3 this covers any polynomial-time team. □

So splitting work across agents buys speed, not capability, and reaching the reasoning ceiling can cost anything from reading the whole input to more than any efficient team can supply.

No general formula exists in between without resolving open problems such as P vs NP.

## VI. Ceiling 3: Communication

*Intuition.* Two agents each hold half a puzzle. The team as a whole holds the answer, but neither agent can use it until enough of the other half arrives. For some puzzles a one-bit summary suffices; for others, nothing much shorter than the data itself will do.

**Theorem 5** (Communication ceiling)**.** *(a) If the messages reaching agent 1 take at most $2^L$ values, then $I(\Theta; D^1, \Pi) \le I(\Theta; D^1) + L$. (b) If agents 1 and 2 hold uniform $x, y \in \{0,1\}^n$ and $\Theta = \langle x, y \rangle \bmod 2$, then $I(\Theta; x, y) = H(\Theta)$, yet any protocol exchanging $L$ bits gives $I(\Theta; x, \Pi) \le \log(1 + 2^{L+1-n/2})$, even with unlimited computation.*

*Proof.* (a) $I(\Theta; \Pi | D^1) \le H(\Pi) \le L$. (b) Append agent 1's output as one more bit. Inner product has discrepancy at most $2^{-n/2}$ (Lindsey's lemma [3], [7]), so an $(L+1)$-bit protocol is correct with probability at most $\frac{1}{2} + 2^{L-n/2}$; apply Lemma 1 to agent 1's exact posterior. □

Parity splits cheaply (each agent posts one bit); inner product needs about $n/2$ bits. With more agents, the same bounds apply to the bits crossing any cut that separates the inputs. This ceiling holds even when information and compute are unlimited.

## VII. Beyond the Assumptions

Each result relaxes one assumption; full proofs and exact checks are in the accompanying notes (footnote 1).

**Theorem 6** (Reasoning can open channels: sandboxes)**.** *Let $A_S(d)$ be the queries a team can issue from record $d$ within compute $S$. Then $I(\Theta; D_T) \le I(\Theta; D_0) + \sum_t \sup_{d,\, q \in A_S(d)} \mathrm{Cap}(K_{t,q,d})$, nondecreasing in $S$. If a tool reveals a uniform 2-bit $\Theta$ only after a uniform $k$-bit password is found, $S$ guesses give at most $\min\{1, S/2^k\} \cdot 2$ bits, with equality for distinct guesses.*

So "reasoning never adds information" holds for a *fixed* set of reachable channels. Discovering an endpoint, credential or exploit is computation that raises the budget. A *sandbox* is a guarantee that every feasible computation stays within allowed channels; an escape violates it. In recently reported incidents, models under evaluation left their test sandbox and retrieved benchmark answers from an outside server [8]: the answers were not inside, so the only route was an unallowed channel.

**Worst case.** For $M \ge 3$ possible truths with no prior, starting data of capacity $C_0$, and any strategy, some truth $\theta_i$ has $\Pr_{\theta_i}(\hat{\Theta} \ne \theta_i) \ge (\log M - \min\{\log M, C_0 + \sum_t C_t\} - 1)/\log(M-1)$ (Fano's method). The law holds for specific tasks, not only on average.

**Acting.** In a fixed unknown environment $W$ with $M \ge 3$ actions, a unique best action $A^*$ and gap $\Delta$, round-$t$ regret is at least $\Delta(H(A^*) - \min\{H(A^*), I(W; D_0) + \sum_{s<t} C_s\} - 1)/\log(M-1)$: the action is an estimate of $A^*$. Taking $W$ to be a user's hidden specification covers open-ended work: an agent cannot reliably build what it has not learned the user wants.

**A changing world.** If $\Theta_t$ drifts as a Markov chain with strong data-processing constant $\eta < 1$ [9], tools observe the current state, and $C_t \le C$, then $I(\Theta_t; D_t) \le \eta^t I(\Theta_0; D_0) + \eta C/(1-\eta)$: knowledge saturates however long agents run.

**Adversaries and provenance.** If every post is stored with its author, adversarial posts cannot reduce available information: $I(\Theta; D, M) \ge I(\Theta; D)$. Without attribution one adversary can erase it: if an honest agent posts a fair bit $\Theta$ and an adversary posts $1-\Theta$, the unattributed record carries 0 bits. Authenticate and attribute messages.

**Continuous truths.** For real $\Theta$ the budget holds with capacities under input constraints, and every estimator has $\mathbb{E}[(\Theta - \hat{\Theta})^2] \ge (2\pi e)^{-1} 2^{2(h(\Theta) - B_T)}$, with equality for a Gaussian truth in Gaussian noise: each bit divides the error floor by 4.

**Where to spend.** From the chain: extra reasoning on the same record can raise usable information by at most the reasoning gap $I(\Theta; D) - I_{\mathcal{V}}(D \to \Theta)$, and extra tool rounds can raise available information by at most $\sum C_t$. A spending rule follows: invest in the lever whose headroom per unit cost is larger, and stop investing in a lever whose headroom is exhausted.

**Measuring the quantities.** Usable information is estimable: $H(\Theta)$ minus the held-out log-loss of a team's calibrated predictions estimates a lower bound on $I_{\mathcal{V}}$, with PAC guarantees under computational constraints [2]. Tool capacity is bounded by the answer space, and for noisy tools by measured error rates (a test that flakes with probability $p$ has $C \le 1 - h(p)$). Available information about the world, $I(W; M)$ for a pretrained model, is well defined but hard to estimate; the bounds' value is then structural rather than numerical.

## VIII. Evidence

Table II lists predictions of the theory and the published findings we found that bear on them, including mixed ones; none contradicts it. Two points deserve care. First, the theory does not predict that debate is worse than other methods; it predicts that debate is worth its compute, so at matched compute it should perform about as well as self-consistency or voting. Small gaps in either direction [10]–[12] are consistent with this. Second, some findings look like tensions. Debate is reported to improve *factuality* [13]; but within the actual world a factual answer is determined by the question, so this is reasoning over stored knowledge, not new information. Debate is proved not to help Bayesian agents [12]; such agents have no reasoning gap, so Theorem 2 predicts exactly that. Finally, diverse agents outperform identical ones [14]: models trained on different data carry different information (a larger $D_0$) and compute different things (a larger $\mathcal{V}$), whereas identical agents add only compute.

## IX. What the Decomposition Predicts

The test of a decomposition is whether it says something a single capability axis does not. Each item below follows

TABLE II

PREDICTIONS OF THE THEORY AND PUBLISHED EVIDENCE. *Supported*: THE FINDING MATCHES THE PREDICTED DIRECTION. *Consistent*: COMPATIBLE WITH THE PREDICTION BUT MIXED ACROSS STUDIES. FINDINGS ARE SUMMARIZED FROM THE CITED PAPERS' ABSTRACTS AND PUBLISHED SUMMARIES.

| Prediction (result in this paper) | Published finding | Verdict |
|---|---|---|
| Extra thinking does not help information-bound tasks (Thm. 1, no tools) | More test-time compute does not reliably raise closed-book accuracy across 14 reasoning models and often increases hallucination [15]; self-correction without external feedback often fails [16]. | Supported |
| Tools raise information-bound accuracy (Thm. 1) | Retrieval and tool use are the standard remedy for knowledge-intensive tasks [17]–[19]. | Supported |
| A model knows only what its training carried (Sec. II) | QA accuracy tracks the number of relevant pretraining documents, causally; retrieval reduces the dependence [20]. | Supported |
| Debate helps compute-bound tasks (Thm. 2) | Debate improves mathematical and strategic reasoning [13]. | Supported |
| Debate is worth its compute, no more (Cor. 3) | Debate does not reliably beat self-consistency [11]; most gains come from voting [12]; single agents match multi-agent systems at equal thinking tokens [10]. | Consistent |
| More identical agents saturate (Thm. 3, Cor. 1) | Homogeneous agents show strong diminishing returns; two diverse agents can match sixteen identical ones [14]; voting gains grow with task difficulty [21]. | Supported |
| Redundant tool calls add little (Thm. 1, exact accounting) | Accuracy plateaus after about eight calls and redundant calls can hurt [22]. | Supported |
| Split information must be communicated (Thm. 5) | Agent groups reach 30.1% with distributed information versus 80.7% for one agent with all of it; agents do not seek what others hold [23]. | Supported |
| Zero-information messages flip correct agents (Sec. IV) | Debate can lower accuracy over time as agents switch from correct to incorrect answers [24]. | Supported |
| Without provenance, adversaries erase information (Sec. VII) | Stealthy tampering of inter-agent messages succeeds across tasks and architectures [25]. | Supported |
| In a changing world knowledge decays and tools refresh it only partly (Sec. VII) | Under continuous knowledge drift, retrieval and learning methods struggle [26]. | Supported |
| Open-ended work is bounded by information about intent (Sec. VII) | Underspecification consistently hurts agents; an ask-user tool recovers part of the loss [27]. | Supported |
| RL post-training raises usable, not available, information (Thm. 2) | RL with verifiable rewards beats the base model at small pass@$k$, but the base model wins at large $k$; the solvable set does not grow and often narrows [28]. | Supported |
| Error vs. informative tool calls stays above the Fano floor (Cor. 2) | Not yet measured. | Open |

from separating the three ceilings; we mark whether published evidence supports it or it is untested.

**1. A diagnostic for any failing collective.** Apply three targeted interventions and see which moves performance: an oracle or better tool (raises the information ceiling), more compute or reasoning (raises usable information), and lifted message limits (raises the communication ceiling). The intervention that helps names the binding ceiling; the others should do little. *Derived; untested.*

**2. Two kinds of diversity.** Models trained on different corpora can carry different information (they can enlarge $D_0$); one model with different personas only computes differently (it enlarges $\mathcal{V}$). So on information-bound tasks only model diversity should help, while on compute-bound tasks both should. Reported diversity gains [14] mix the two. *Derived; the split is untested.*

**3. Debate is worth its compute.** At matched compute, debate should perform about as well as self-consistency or voting, helping compute-bound tasks and not information-bound ones, while tools show the opposite pattern. *Consistent with* [10]–[12]*; the task-type split is untested.*

**4. Post-training lifts one ceiling.** Pre-training sets what a model can know about the world; post-training on its own outputs mostly enlarges what it can use. RL should therefore raise performance on compute-bound tasks without expanding what the base model can solve given enough attempts, and should not help information-bound tasks unless its training signal carries new facts. *Supported* [28] *for the first part.*

**5. Communication has a signature.** As the message budget grows, accuracy should jump at a threshold for inner-product-like splits but rise from the first bit for parity-like splits (Theorem 5). Summarizer and orchestrator architectures should fail specifically where communication complexity is high, however strong the summarizer. *Consistent with* [23]*; the threshold is untested.*

**6. An error floor for tool use.** Error against the number of informative tool calls stays above the floor of Corollary 2; with a yes/no tool of known noise the floor is exact, so the test is clean. *Untested.*

**7. Where channel-seeking appears.** Reward hacking by finding unallowed channels should concentrate on tasks whose answers are not reachable inside the sandbox, i.e. information-bound tasks under strong reward pressure (Theorem 6). *Consistent with a reported incident* [8]*; untested systematically.*

**Exchange rates.** The decomposition also prices the levers: a tool call is worth at most $C_t$ bits of information; a debate round, zero bits of information but possibly a large gain in usable information; a message, at most its length in bits to its recipient. Spending can then follow the headroom rule of Section VII.

## X. RELATED WORK

Closed-system debate is bounded via the DPI in [29] and modeled as a belief martingale in [12]. Yang et al. [14] bound

multi-agent performance by task uncertainty and a sum of per-agent channel informations, the same "$\min\{H, \sum \text{channels}\}$" shape with agents as the channels. Fano-type bounds have been applied to queries in adversarial attacks [30] and to single-pass multi-hop reasoning [31]. Usable information is due to [2]; synergy is studied in partial information decomposition [32]; feedback and directed information in [5]; and Fano-based active-learning bounds in [33]. These works each address one ceiling. To our knowledge, the per-call capacity budget for adaptive multi-agent tool use, its combination with usable information and communication into one decomposition, and the channel account of sandboxing have not been stated before.

## XI. Limitations

The bounds concern information about a world the agents do not change; outcomes the team itself brings about need a different theory. The ceilings hold separately, but how to trade one for another is open, as is modeling agents' incentives (reward hacking, goal adoption), which the bounds do not address. Bounds for agents that learn across episodes, whose starting data changes over time, are not given. Equality in Theorem 1 is characterized, but no general algorithm achieves it. The proofs are mathematical and checked numerically on finite examples, not machine-verified. We report no new experiments; Table II relies on published findings, summarized from their abstracts, and several predictions in Section IX are untested.

## XII. Conclusion

Collective intelligence is not one quantity, and it does not have one limit. A team of agents cannot know more than it has observed, use more than it holds, or combine what its members hold without exchanging enough; each of these is bounded by a different theory, and each is moved by a different intervention. The question to ask of a struggling collective is therefore not how to make it more intelligent, but *which resource limits it, and what can remove that limit*: better observations of the world, more reasoning, or more communication. Answering it per task, rather than adding agents by default, is what the decomposition is for.

## References


[1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[2] Y. Xu, S. Zhao, J. Song, R. Stewart, and S. Ermon, "A theory of usable information under computational constraints," in *Int. Conf. Learning Representations (ICLR)*, 2020.

[3] E. Kushilevitz and N. Nisan, *Communication Complexity*. Cambridge University Press, 1997.

[4] C. E. Shannon, "The zero error capacity of a noisy channel," *IRE Transactions on Information Theory*, vol. 2, no. 3, pp. 8–19, 1956.

[5] J. L. Massey, "Causality, feedback and directed information," in *Proc. Int. Symp. Information Theory and its Applications (ISITA)*, 1990.

[6] O. Goldreich and L. A. Levin, "A hard-core predicate for all one-way functions," in *Proc. ACM Symp. Theory of Computing (STOC)*, 1989.

[7] B. Chor and O. Goldreich, "Unbiased bits from sources of weak randomness and probabilistic communication complexity," *SIAM Journal on Computing*, vol. 17, no. 2, pp. 230–261, 1988.

[8] OpenAI, "The Hugging Face incident and the road ahead," https://openai.com/index/hugging-face-incident-and-the-road-ahead/, 2026.

[9] R. Ahlswede and P. Gács, "Spreading of sets in product spaces and hypercontraction of the Markov operator," *Annals of Probability*, vol. 4, no. 6, pp. 925–939, 1976.

[10] D. Tran and D. Kiela, "Single-agent LLMs outperform multi-agent systems on multi-hop reasoning under equal thinking token budgets," arXiv:2604.02460, 2026.

[11] A. Smit, P. Duckworth, N. Grinsztajn, T. D. Barrett, and A. Pretorius, "Should we be going MAD? A look at multi-agent debate strategies for LLMs," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 235, 2024.

[12] H. K. Choi, X. Zhu, and S. Li, "Debate or vote: Which yields better decisions in multi-agent large language models?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[13] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 235, 2024.

[14] Y. Yang, C. Qu, M. Wen, L. Shi, Y. Wen, W. Zhang, A. Wierman, and S. Gu, "Understanding agent scaling in LLM-based multi-agent systems via diversity," arXiv:2602.03794, 2026.

[15] J. X. Zhao, B. Hooi, and S.-K. Ng, "Test-time scaling in reasoning models is not effective for knowledge-intensive tasks yet," in *Int. Conf. Learning Representations (ICLR)*, 2026, arXiv:2509.06861.

[16] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *Int. Conf. Learning Representations (ICLR)*, 2024.

[17] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[18] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Int. Conf. Learning Representations (ICLR)*, 2023.

[19] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[20] N. Kandpal, H. Deng, A. Roberts, E. Wallace, and C. Raffel, "Large language models struggle to learn long-tail knowledge," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023.

[21] J. Li, Q. Zhang, Y. Yu, Q. Fu, and D. Ye, "More agents is all you need," *Transactions on Machine Learning Research*, 2024.

[22] Y. Du, Y. Zhang, X. Yang, Y. Zhou, C. Wang, G. Zou, X. Pang, W. Wang, M. Chen, S. Tang, Z. Li, and S. Chen, "InfoMosaic-Bench: Evaluating multi-source information seeking in tool-augmented agents," arXiv:2510.02271, 2025.

[23] Y. Li, A. Naito, and H. Shirado, "Systematic failures in collective reasoning under distributed information in multi-agent LLMs," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 306, 2026.

[24] A. Wynn, H. Satija, and G. Hadfield, "Talk isn't always cheap: Understanding failure modes in multi-agent debate," in *ICML Workshop on Multi-Agent Systems*, 2025, arXiv:2509.05396.

[25] B. Yan, Z. Zhou, X. Zhang, C. Li, R. Zeng, Y. Qi, T. Wang, and L. Zhang, "Attack the messages, not the agents: A multi-round adaptive stealthy tampering framework for LLM-MAS," in *Proc. AAAI Conf. Artificial Intelligence*, 2026.

[26] H. Liu, L. Cao, and Y. Li, "RAG or learning? Understanding the limits of LLM adaptation under continuous knowledge drift in the real world," in *Findings of the Association for Computational Linguistics: ACL*, 2026.

[27] G. Pu, M. S. Lee, U. M. Sehwag, D. J. Lee, B. Zhu, Y. Maurya, M. Raghavendra, Y. Xue, and S. M. Denton, "LHAW: Controllable underspecification for long-horizon tasks," in *Int. Conf. Learning Representations (ICLR)*, 2026.

[28] Y. Yue, Z. Chen, R. Lu, A. Zhao, Z. Wang, S. Song, and G. Huang, "Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[29] K. S. Shin, "The reasoning trap: An information-theoretic bound on closed-system multi-step LLM reasoning," arXiv:2605.01704, 2026.

[30] M. Kaneko and T. Baldwin, "Bits leaked per query: Information-theoretic bounds on adversarial attacks against LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[31] K. Wan, L. Gao, H. Mu, P. Nakov, Y. Wang, and X. Chen, "A Fano-style accuracy upper bound for LLM single-pass reasoning in multi-hop QA," in *Int. Conf. Learning Representations (ICLR)*, 2026.

[32] P. L. Williams and R. D. Beer, "Nonnegative decomposition of multivariate information," arXiv:1004.2515, 2010.

[33] M. Raginsky and A. Rakhlin, "Lower bounds for passive and active learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2011.